

Environmental Pollution Incidents across England (2000-2026)

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```
options(repos = c(CRAN = "https://cloud.r-project.org/"))
library(tidyverse)
library(rio)
library(readr)
library(sf)
#library(lubridate)
```

Data source: <https://www.data.gov.uk/dataset/c8625e18-c329-4032-b4c7-444b33af6780/environmental-pollution-incidents-category-1-and-21>

```
# url
url <- "https://environment.data.gov.uk/api/file/download?fileDataSetId=a5a11813-517a-478b-b"

# download file
download.file(url, destfile = "data/pollution.zip", mode = "wb")

#unzip
unzip("data/pollution.zip",
      files = c("EP_Incidents_Nirs2.csv", "EP_Pollutants_Nirs2.csv"),
      exdir = "data")

# assign to variables
incidents <- rio::import("data/EP_Incidents_Nirs2.csv")
pollutants <- rio::import("data/EP_Pollutants_Nirs2.csv")

# date, year and month
incidents$NOT_DATE <- as.POSIXct(incidents$NOT_DATE, format = "%d/%m/%Y %H:%M")
incidents$YEAR <- as.numeric(format(incidents$NOT_DATE, "%Y"))
```

```

incidents$MONTH <- as.numeric(format(incidents$NOT_DATE, "%m"))

# Convert to spatial object (British National Grid)
df <- st_as_sf(incidents, coords = c("X_CONF", "Y_CONF"), crs = 27700)

# join datasets
df <- df %>%
  left_join(
    pollutants %>% select(NOT_ID, POLL_TYPE, POLLUTANT),
    by = "NOT_ID")

# nominal categorical as factors
df$REGION_WM <- as.factor(df$REGION_WM)
df$AREA_WM <- as.factor(df$AREA_WM)
df$REGION_PF <- as.factor(df$REGION_PF)
df$AREA_PF <- as.factor(df$AREA_PF)
df$COUNTY <- as.factor(df$COUNTY)
df$UNITARY <- as.factor(df$UNITARY)
df$DISTRICT <- as.factor(df$DISTRICT)
df$EP_INC <- as.factor(df$EP_INC)
df$EIL_AIR <- as.factor(df$EIL_AIR)
df$EIL_LAND <- as.factor(df$EIL_LAND)
df$EIL_WATER <- as.factor(df$EIL_WATER)
df$POLL_TYPE <- as.factor(df$POLL_TYPE)
df$POLLUTANT <- as.factor(df$POLLUTANT)

# ordinal categorical as factors
df <- df %>%
  mutate(YEAR = factor(YEAR, levels = sort(unique(YEAR))))
df <- df %>%
  mutate(MONTH = factor(MONTH, levels = sort(unique(MONTH))))

# view variables and observations
str(df)

```

Classes 'sf' and 'data.frame': 20053 obs. of 19 variables:

```

$ NOT_ID   : int  102 299 340 371 418 494 587 664 692 697 ...
$ NOT_DATE : POSIXct, format: "2001-03-08 09:10:00" "2001-03-14 10:40:00" ...
$ REGION_WM: Factor w/ 10 levels "", "Anglian Region",...: 6 6 6 6 6 6 8 6 8 8 ...
$ AREA_WM  : Factor w/ 20 levels "", "Cumbria and Lancashire",...: 2 2 2 7 2 2 19 7 3 19 ...
$ REGION_PF: Factor w/ 9 levels "", "Anglian Region",...: 5 5 5 5 5 5 7 5 7 7 ...
$ AREA_PF  : Factor w/ 23 levels "", "Central", "Cumbria and Lancashire",...: 3 3 7 7 7 3 20 7

```

```

$ COUNTY : Factor w/ 35 levels "", "BEDFORDSHIRE COUNTY",...: 7 18 1 1 1 18 10 5 6 10 ...
$ UNITARY : Factor w/ 47 levels "", "BATH AND NORTH EAST SOMERSET",...: 1 1 1 1 1 1 1 1 1 1 ...
$ DISTRICT : Factor w/ 307 levels "", "ADUR DISTRICT",...: 46 38 297 299 297 291 296 152 45 1 ...
$ NGR_CONF : chr "NY 3634 6325" "SD 82243 31457" "SD 61027 06322" "SJ 23570 86666" ...
$ EP_INC : Factor w/ 1 level "Yes": 1 1 1 1 1 1 1 1 1 1 ...
$ EIL_AIR : Factor w/ 4 levels "Category 1 (Major)",...: 2 4 2 4 2 4 2 4 4 4 ...
$ EIL_LAND : Factor w/ 4 levels "Category 1 (Major)",...: 4 4 4 4 4 4 2 2 4 4 ...
$ EIL_WATER: Factor w/ 4 levels "Category 1 (Major)",...: 4 2 4 2 4 2 3 3 2 1 ...
$ YEAR : Factor w/ 27 levels "2000", "2001",...: 2 2 2 2 2 2 2 2 2 2 ...
$ MONTH : Factor w/ 12 levels "1", "2", "3", "4",...: 3 3 3 3 3 3 3 3 3 3 ...
$ POLL_TYPE: Factor w/ 12 levels "Agricultural Materials and Wastes",...: 2 7 2 1 2 6 3 5 7 5 ...
$ POLLUTANT: Factor w/ 114 levels "Acids", "Adhesives",...: 70 45 70 68 70 75 41 21 45 74 ...
$ geometry :sf POINT of length 20053; first list element: 'XY' num 336340 563250
- attr(*, "sf_column")= chr "geometry"
- attr(*, "agr")= Factor w/ 3 levels "constant", "aggregate",...: NA NA NA NA NA NA NA NA NA NA ...
..- attr(*, "names")= chr [1:18] "NOT_ID" "NOT_DATE" "REGION_WM" "AREA_WM" ...

```

```
summary(df)
```

NOT_ID		NOT_DATE		REGION_WM	
Min. :	102	Min. :	2000-02-15 15:45:00	North East Region:3949	
1st Qu.:	274420	1st Qu.:	2004-10-26 16:41:00	North West Region:3132	
Median :	820637	Median :	2010-09-08 12:32:00	Midlands Region :3083	
Mean :	979622	Mean :	2011-11-01 19:43:23	South West Region:2767	
3rd Qu.:	1637339	3rd Qu.:	2018-07-26 13:21:00	Anglian Region :2660	
Max. :	2480368	Max. :	2026-03-20 11:39:00	South East Region:2040	
				(Other) :2422	

	AREA_WM	REGION_PF
Yorkshire	:2953	North East Region:3907
Cumbria and Lancashire	:2016	Midlands Region :3230
Wessex	:1430	North West Region:3030
Herts and North London	:1370	South West Region:2776
Kent, South London and East Sussex	:1359	Anglian Region :2748
Devon, Cornwall and the Isles of Scilly:	1339	South East Region:2020
(Other)	:9586	(Other) :2342

	AREA_PF	COUNTY
Yorkshire	:2988	:7205
West Midlands	:2168	LANCASHIRE COUNTY:1159
Cumbria and Lancashire:	1859	KENT COUNTY : 738
East Anglia	:1741	DEVON COUNTY : 656
North East	:1574	CUMBRIA COUNTY : 611
Wessex	:1502	CORNWALL COUNTY : 588

(Other)	:8221	(Other)	:9096
	UNITARY		DISTRICT
	:17745		: 2317
EAST RIDING OF YORKSHIRE:	219	LEEDS DISTRICT	: 463
COUNTY OF HEREFORDSHIRE :	160	KIRKLEES DISTRICT :	311
NORTH LINCOLNSHIRE :	129	WAKEFIELD DISTRICT :	263
SOUTH GLOUCESTERSHIRE :	93	BRADFORD DISTRICT :	256
BLACKBURN :	87	CALDERDALE DISTRICT:	202
(Other)	: 1620	(Other)	:16241
NGR_CONF	EP_INC	EIL_AIR	
Length:20053	Yes:20053	Category 1 (Major)	: 288
Class :character		Category 2 (Significant):	4175
Mode :character		Category 3 (Minor)	: 2797
		Category 4 (No Impact)	:12793

	EIL_LAND		EIL_WATER
Category 1 (Major)	: 248	Category 1 (Major)	:1478
Category 2 (Significant):	6104	Category 2 (Significant):	9315
Category 3 (Minor)	: 2813	Category 3 (Minor)	:2076
Category 4 (No Impact)	:10888	Category 4 (No Impact)	:7184

YEAR		MONTH		POLL_TYPE
2002	: 1453	7	:2289	Atmospheric Pollutants and Effects:2931
2001	: 1292	8	:2124	Sewage Materials :2846
2003	: 1245	6	:1999	Specific Waste Materials :2637
2004	: 1222	9	:1818	Oils and Fuel :2092
2005	: 1021	5	:1725	Inert Materials and Wastes :1676
2006	: 937	4	:1641	Agricultural Materials and Wastes :1535
(Other):12883		(Other):8457		(Other) :6336
			POLLUTANT	geometry
Crude Sewage			: 1911	POINT :20053
Not Identified			: 1534	epsg:27700 : 0
Slurry and Dilute Slurry			: 948	+proj=tmer...: 0
Construction and Demolition Materials and Wastes:			877	
Smoke			: 768	
Diesel			: 646	
(Other)			:13369	

```

# plot map, facet by year
ggplot(df) +
  geom_sf(aes(color = factor(EIL_WATER))) +
  facet_wrap(~ YEAR) +
  labs(
    x = "Longitude",
    y = "Latitude",
    color = "Environmental Impact Level (Water)",
    title = "Map of incidents by category (2000-2026)"
  ) +
  theme_bw() +
  theme(
    legend.position = "right"
  )

```



Figure 1: Map of pollution incidents by category (2000-2026)

```
# create data subset for 2025 only
df_25 <- df %>%
  filter(YEAR == 2025)
```

```
# plot 2025 only as a map
ggplot(df_25) +
  geom_sf(aes(color = factor(EIL_WATER))) +
```

```
facet_wrap(~ YEAR) +  
labs(  
  x = "Longitude",  
  y = "Latitude",  
  color = "Environmental Impact Level (Water)",  
  title = "Map of incidents by category (2025)"  
) +  
theme_bw() +  
theme(  
  legend.position = "right"  
)
```

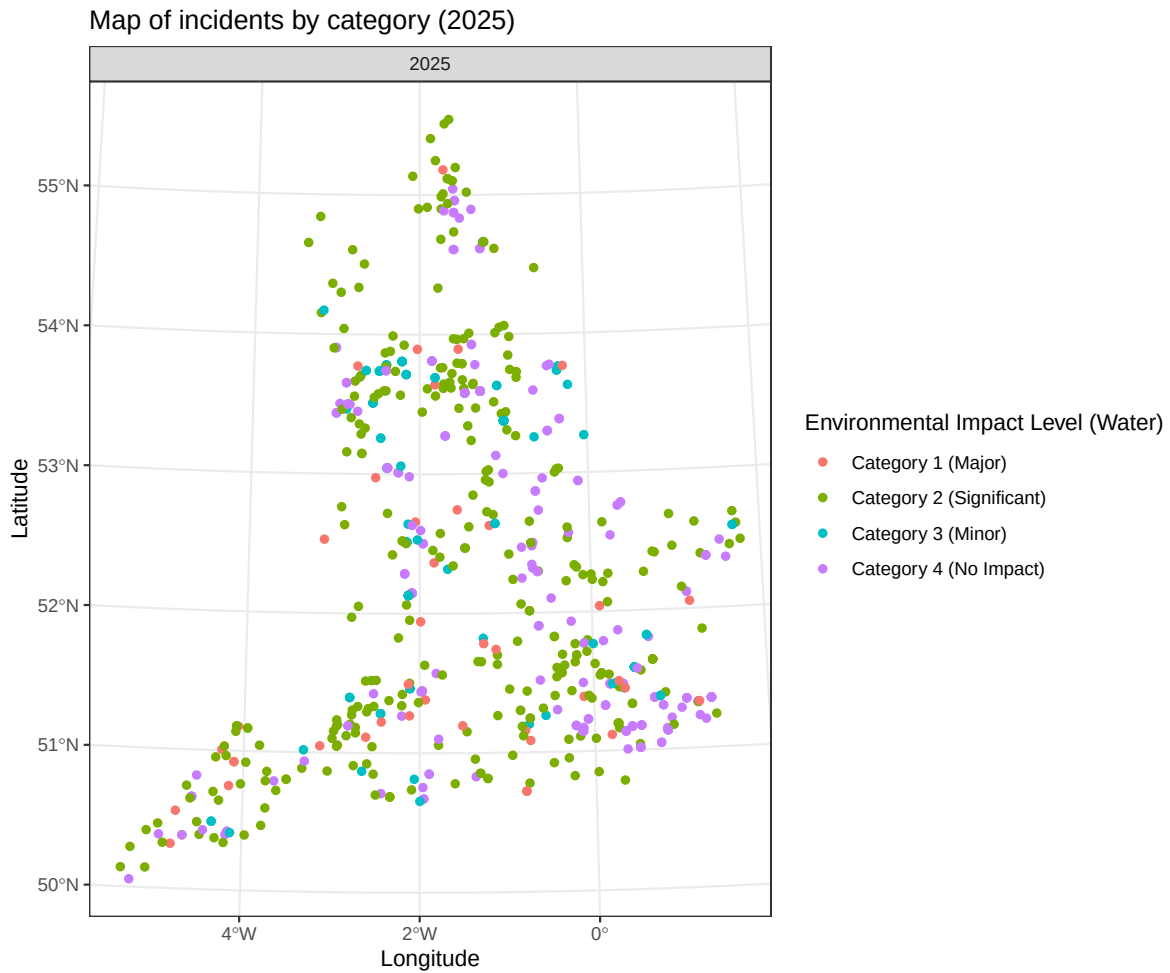


Figure 2: Map of pollution incidents by category (2025)

```
# plot 2025 only as a map, facet by incident category
ggplot(df_25) +
  geom_sf(aes(color = factor(POLL_TYPE))) +
  facet_wrap(~ EIL_WATER) +
  labs(
    x = "Longitude",
```



```
y = "Latitude",  
color = "Pollutant Type",  
title = "Map of incidents by category and pollutant type (2025)"  
) +  
theme_bw() +  
theme(  
  legend.position = "right"  
)
```

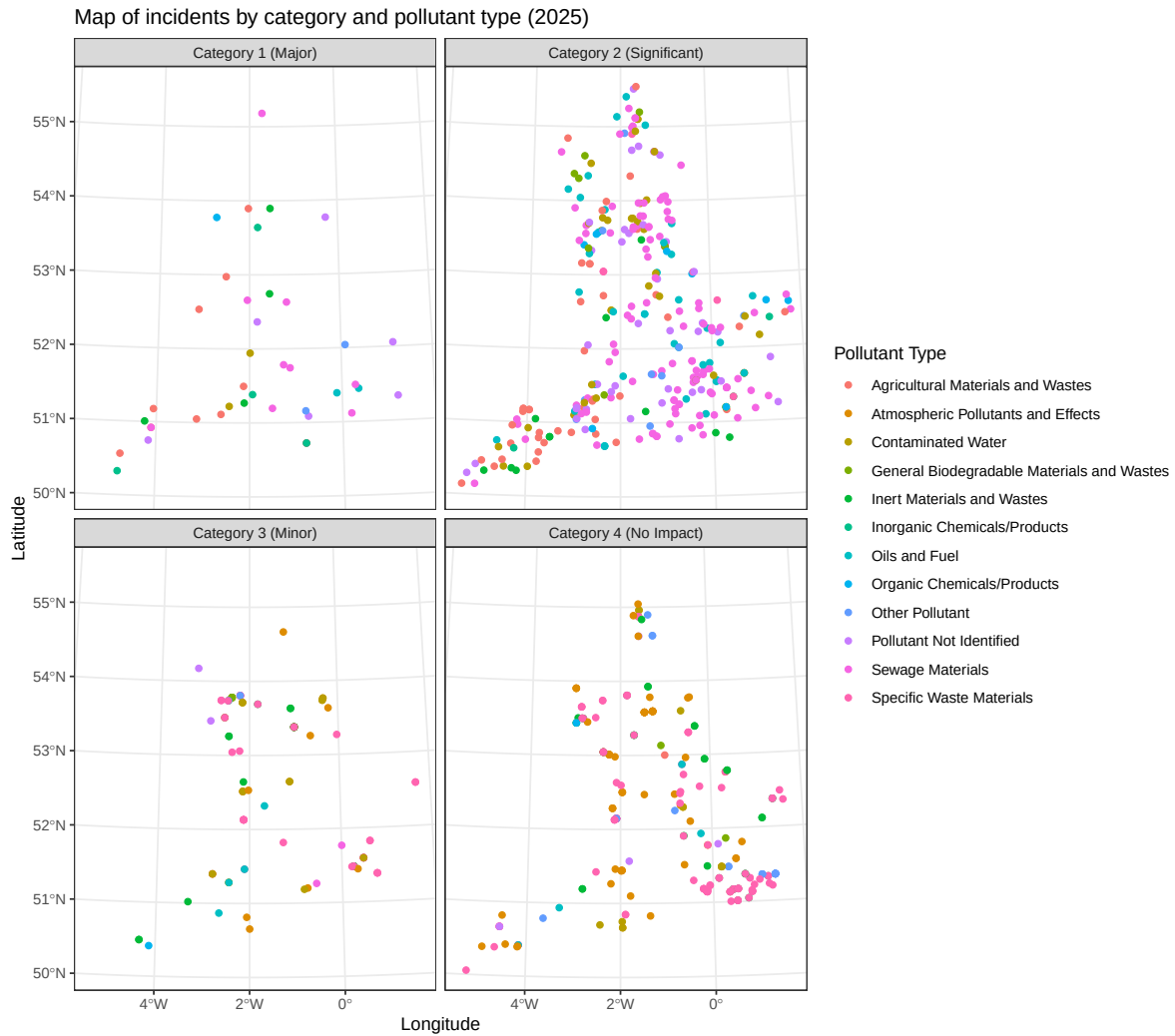


Figure 3: Map of pollution incidents by pollutant type and category (2025)

```
# Region and Category - 27 years (2020 - 2026)
ggplot(df,
  aes(x = EIL_WATER, fill = EIL_WATER)) +
  geom_bar() +
  facet_wrap(~ REGION_PF,
    nrow = 3, ncol = 3,
```

```

        labeller = label_wrap_gen(width = 15)) +
labs(
  x = NULL,
  y = "Counts",
  fill = "Category",
  title = "Frequencies of incidents by Region and Category (2020-2026)"
) +
theme_bw() +
theme(
  axis.text.x = element_blank(),
  axis.ticks.x = element_blank(),
  legend.position = "right"
)

```

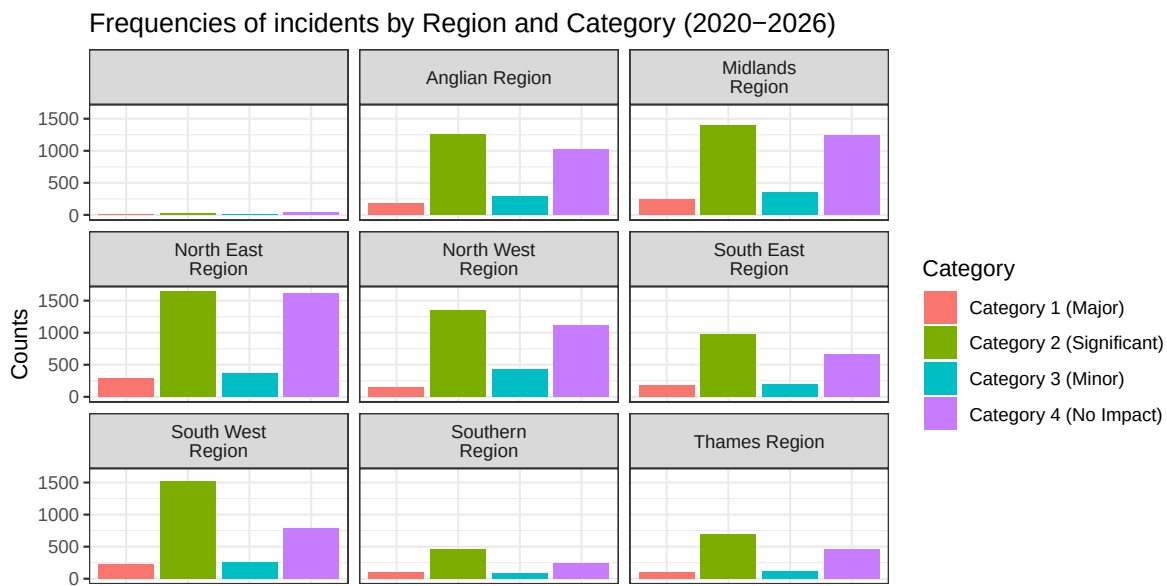


Figure 4: Bar chart of incidents by region and category (2020-2026)

```

# Region and Category - 1 year (2025)
ggplot(df_25,
  aes(x = EIL_WATER, fill = EIL_WATER)) +
geom_bar() +
facet_wrap(~ REGION_PF,
  nrow = 3, ncol = 3,
  labeller = label_wrap_gen(width = 15)) +
labs(

```

```

x = NULL,
y = "Counts",
fill = "Category",
title = "Frequencies of incidents by Region and Category (2025)"
) +
theme_bw() +
theme(
  axis.text.x = element_blank(),
  axis.ticks.x = element_blank(),
  legend.position = "right"
)

```

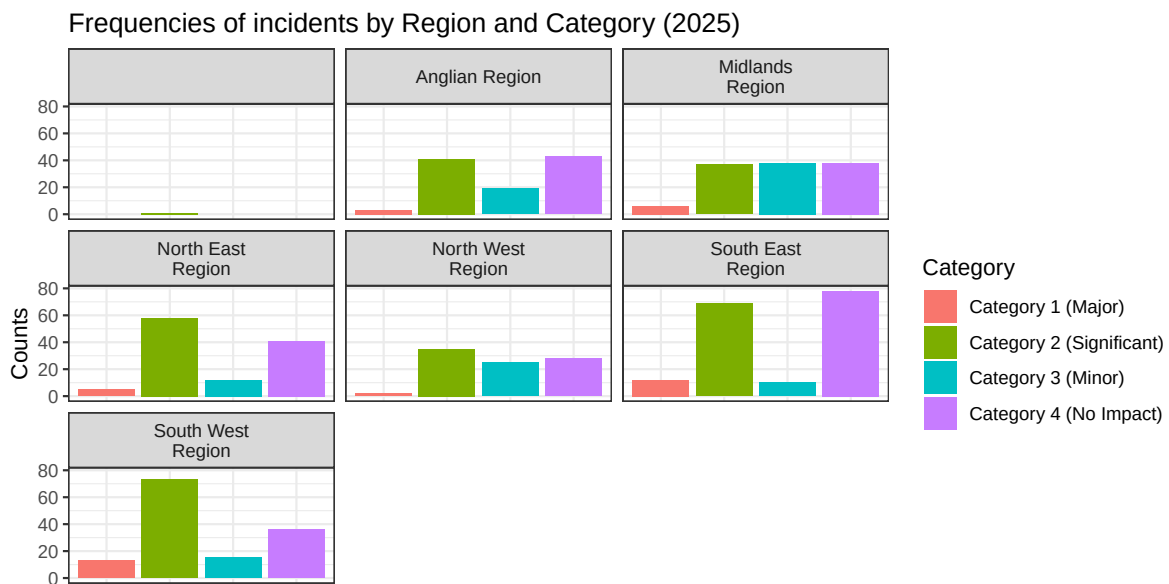


Figure 5: Bar chart of incidents by region and category (2025)

```

# Category and Pollution - 27 years (2020 - 2026)
ggplot(df,
  aes(x = POLL_TYPE, fill = POLL_TYPE)) +
geom_bar() +
facet_wrap(~ EIL_WATER) +
labs(
  x = NULL,
  y = "Counts",
  fill = "Category",
  title = "Frequencies of incidents by Category and Pollution Type (2020-2026)"
)

```

```
) +
theme_bw() +
theme(
  axis.text.x = element_blank(),
  axis.ticks.x = element_blank(),
  legend.position = "right"
)
```

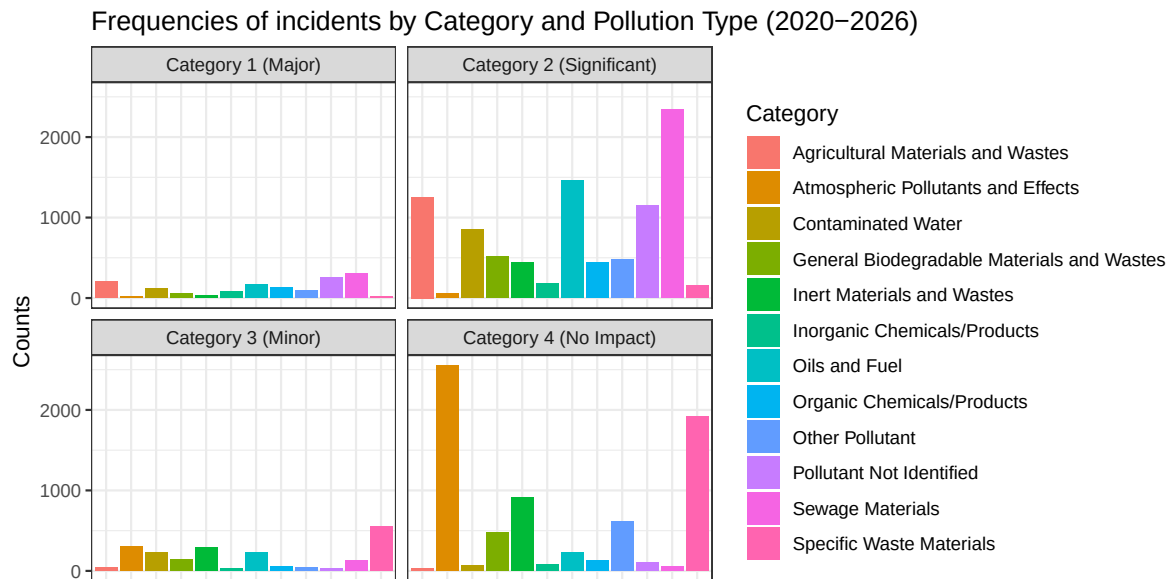


Figure 6: Bar chart of incidents by category and pollutant type (2000-2026)

```
# Category and Pollution - 1 year (2025)
ggplot(df_25,
  aes(x = POLL_TYPE, fill = POLL_TYPE)) +
geom_bar() +
facet_wrap(~ EIL_WATER) +
labs(
  x = NULL,
  y = "Counts",
  fill = "Category",
  title = "Frequencies of incidents by Category and Pollution Type (2025)"
) +
theme_bw() +
theme(
  axis.text.x = element_blank(),
```

```
axis.ticks.x = element_blank(),
legend.position = "right"
)
```

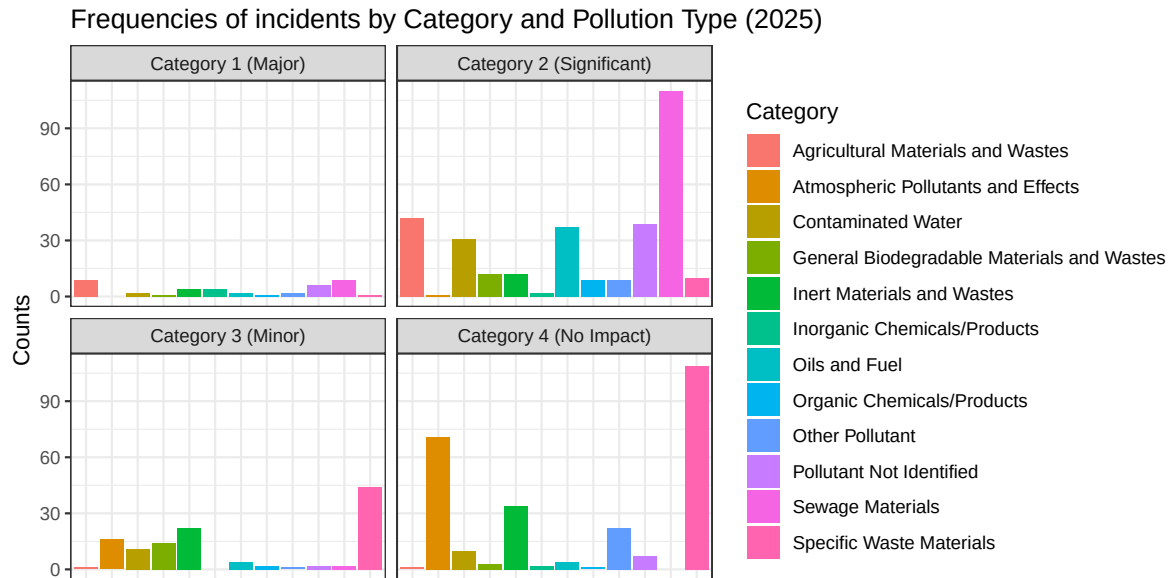


Figure 7: Bar chart of incidents by category and pollutant type (2025)